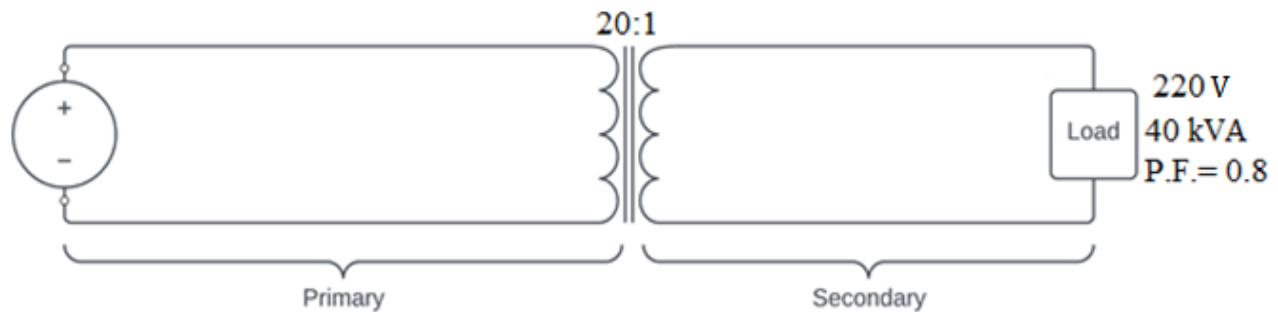

Final Exam (11th Dec 2023) – ECE 3113 Energy Conversion

- Make sure that you fill in your OU ID # and name so that your attendance will be checked.
 - Please write your answers neatly and legibly.
 - You can use the reverse side of sheets for solving procedure.
 - Your answers are required to be shown to three significant digits.
 - Please show all your work to make it clear how you arrived at your solution.
-

Problem 1 (20 Points)

Consider a single-phase transformer with a capacity of 50 kVA, and turn ratio of 20: 1. The system frequency is 60 Hz . A load, which is connected to the secondary winding of the transformer, absorbs the total 40 kVA power at 0.8 pf (*lagging*) and operates at 220 V (RMS). Assuming the transformer is an ideal one, find the following:

- Primary supply voltage.
- Load impedance.
- Load impedance referred to the primary.
- The real and reactive power supplied to the primary winding.



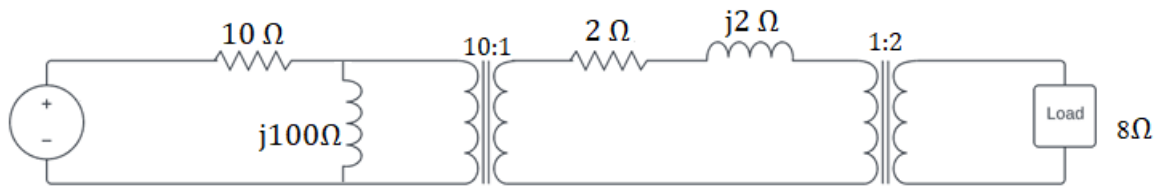
Problem 2 (20 Points)

For a motor with the following characteristics, $R_a = 0.35 \, \Omega$, $V_{dc} = 110 \, \text{V}$, $V_{brush} = 0 \, \text{V}$, no-load speed $n_0 = 1000 \, \text{turns/minute}$,

- a) Derive the equation of the output curve (i.e., the n-Torque curve).
- b) Draw the associated Speed-Torque output curve.

Problem 3 (20 points)

Given the parameters in the circuit and supply input voltage of $v_{in}(t) = 110 \sin(40\pi t)$ determine the output voltage across the load.



$$v_{in}(t) = 110 \sin(40\pi t)$$

Problem 4 (20 points)

- a) Derive the mathematical equations of rotating magnetic field density (B) for given three phase AC currents, $i_a(t)$, $i_b(t)$ and $i_c(t)$. Consider the three phase windings to be spaced 120° apart. Assuming linear magnetism ($B = KI$).

$$i_a(t) = I_m \sin(\omega t + 30^\circ)$$

$$i_b(t) = I_m \sin(\omega t - 90^\circ)$$

$$i_c(t) = I_m \sin(\omega t - 210^\circ)$$

- b) Describe and explain the conditions for generation of rotating magnetic field.

Problem 5 (20 points)

A DC motor with parameters, $R_{am} = 0.2 \Omega$, $V_{dc} = 100 V$, $V_{brush} = 0 V$ has no-load speed of $n_0 = 1700 \text{ turns/minute}$.

- Calculate the maximum output torque T_{max} and maximum current $I_{am,max}$?
- What is the output torque T_{out} , current I_{am} , and efficiency of this motor when $n = 1000 \text{ turns/minute}$?

